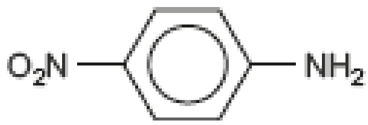
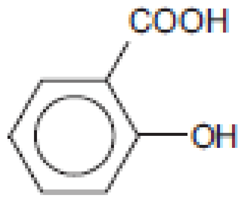
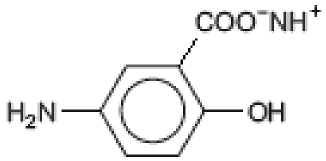
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10.	(a)	(i)		 (1)  (1)		2		2		
		(ii)		the wavelength of the light absorbed increases as the pH increases; in the visible spectrum, violet has the shortest wavelength and this increases as the colour moves towards red			1	1		
		(iii)		$f = c / \lambda$ $f = 3.00 \times 10^{8} / 385 \times 10^{-9}$ (1) $f = 7.79 \times 10^{14}$ (1)	1	1		2	2	
		(iv)		$E = hf$ $E = 6.63 \times 10^{-34} \times 7.79 \times 10^{14}$ (1) $E = 5.16 \times 10^{-19}$ (1) $E = 5.16 \times 10^{-19} \times 6.02 \times 10^{23} = 310684\ \text{J mol}^{-1} = 310.68\ \text{kJ mol}^{-1}$ 311 (1) ecf from part (ii)		3		3	1 1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		tin (or iron) and concentrated hydrochloric acid	1			1		1
		(ii)		there are two signals of equal size / area (1) the 4 aromatic protons are in identical environments and give singlet / no splitting (1) the 4 NH_2 protons are in identical environments and give singlet / no splitting (1)			3	3		
		(iii)				1		1		
		(iv)				1		1		
				Question 10 total	**2**	**8**	**4**	**14**	**4**	**1**